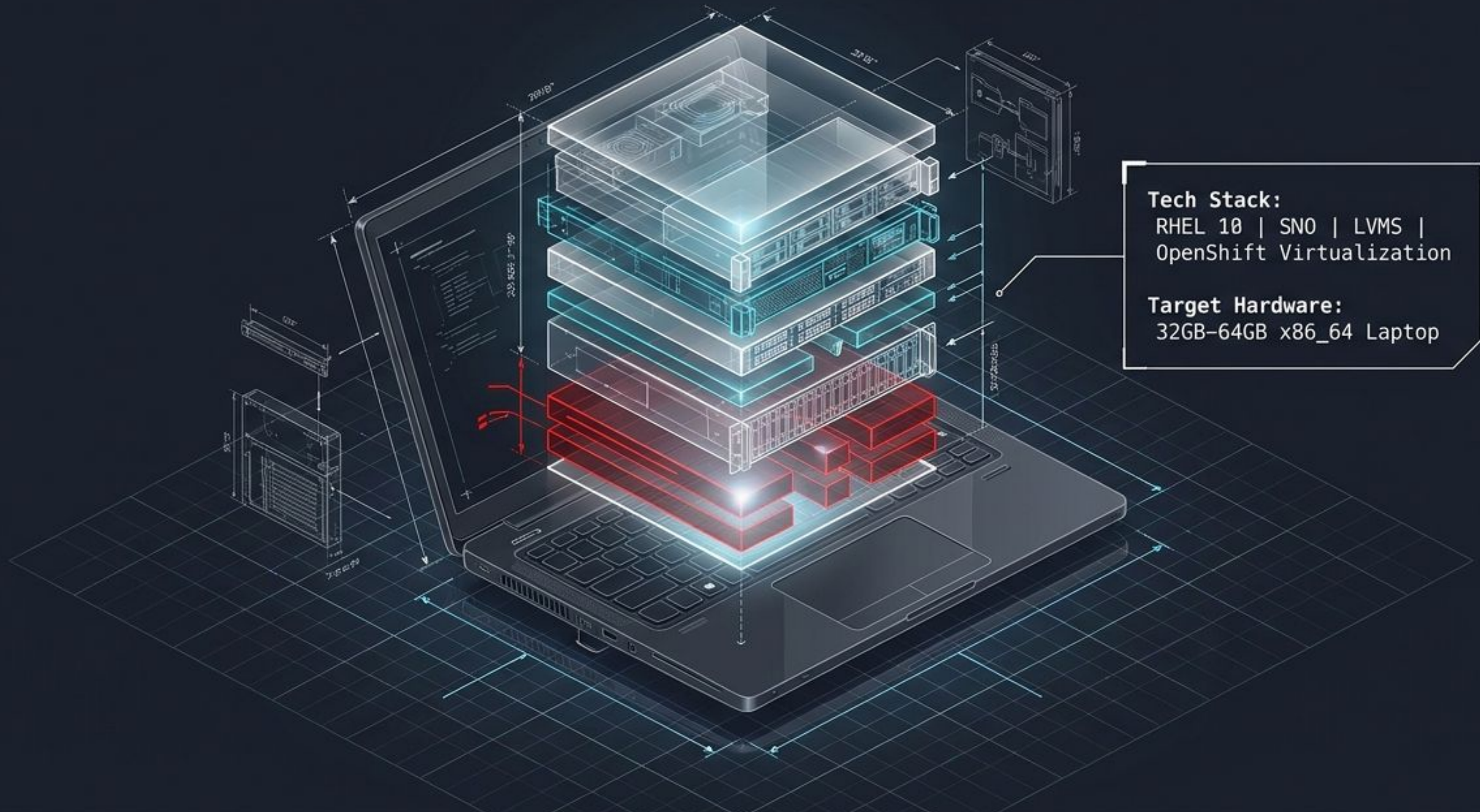


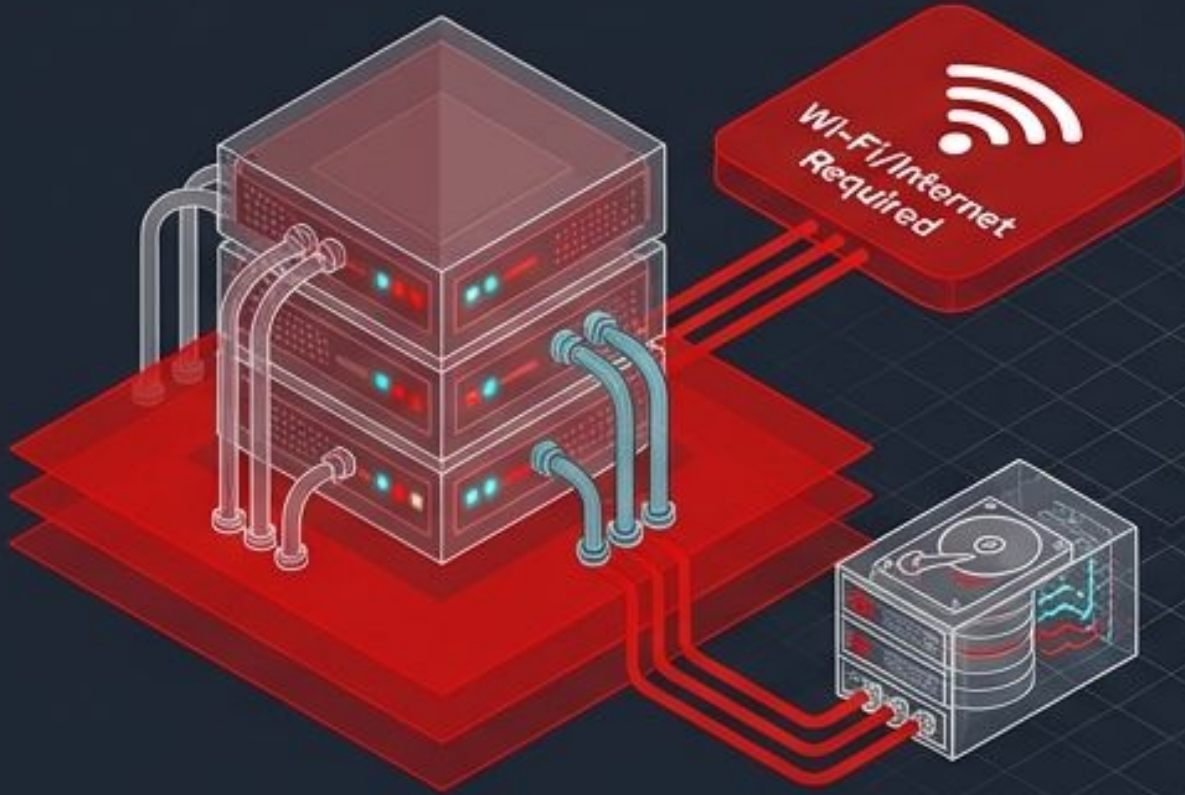
Datacenter in a Backpack

Architecting a Single-Laptop OpenShift Virtualization Demo Environment



The Field Demo Conundrum

The Enterprise Standard



- Highly Available (HA) Control Planes
- Dedicated Ceph/ODF Storage
- Bare Metal Hypervisors
- Requires constant connectivity

The Laptop Constraint



- Constrained RAM/CPU
- Single physical NVMe drive
- Air-gapped/offline operation
- Requires a nested hypervisor (VM inside a Pod inside a VM/Host)

Solution: A highly tuned, nested virtualization stack running Red Hat Single-Node OpenShift (SNO).

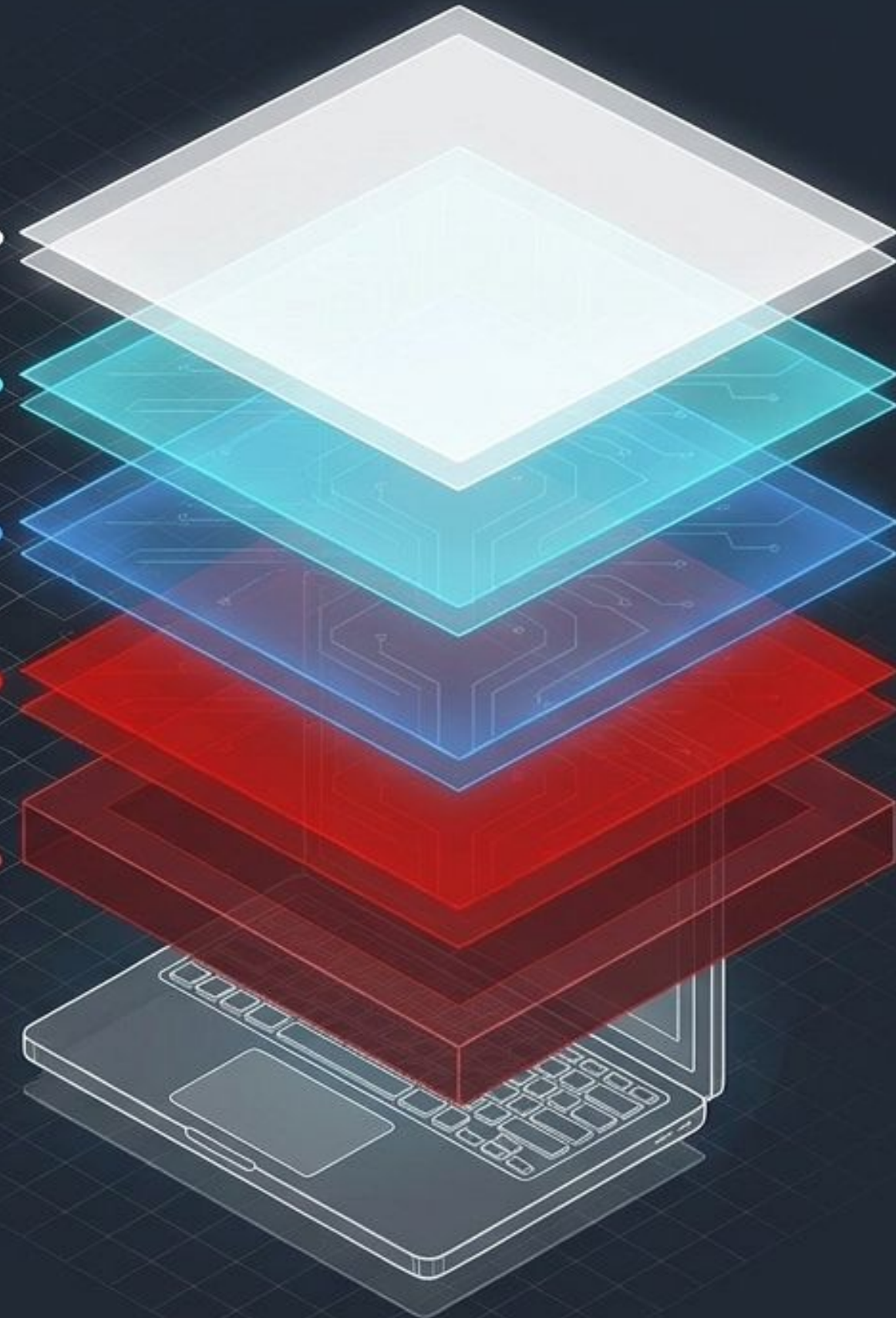
Layer 5: Nested Virtual Machine
Workloads (e.g., RHEL 9 Guest)

Layer 4: OpenShift Virtualization
(CNV / KVM / QEMU)

Layer 3: LVM Storage Operator (LVMS)
& Multus Networking

Layer 2: Single-Node OpenShift
(SNO / RHCOS)

Layer 1: Physical Hardware &
RHEL 10 Host OS



Layer 1: The RHEL 10 Foundation

Minimum Physical Foundation



- **Processor:** Intel VT-x or AMD-V enabled CPU



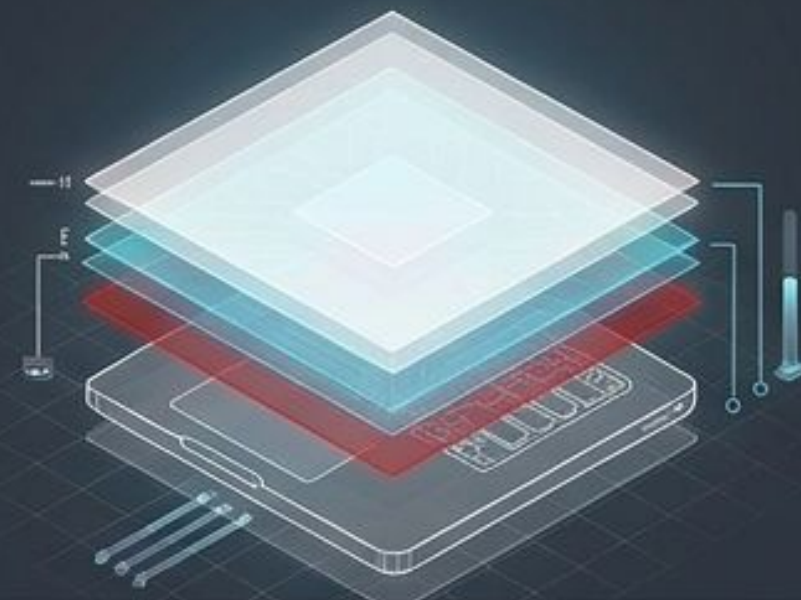
- **Memory:** 64GB RAM recommended (32GB strict minimum)



- **Storage:** Fast NVMe SSD (minimum 500GB dedicated partition)

RHEL 10

Provides the stable, highly optimized type-1 KVM hypervisor foundation. RHEL 10 supports instance types and preferences directly impacting virtualization performance.



Host Preparation

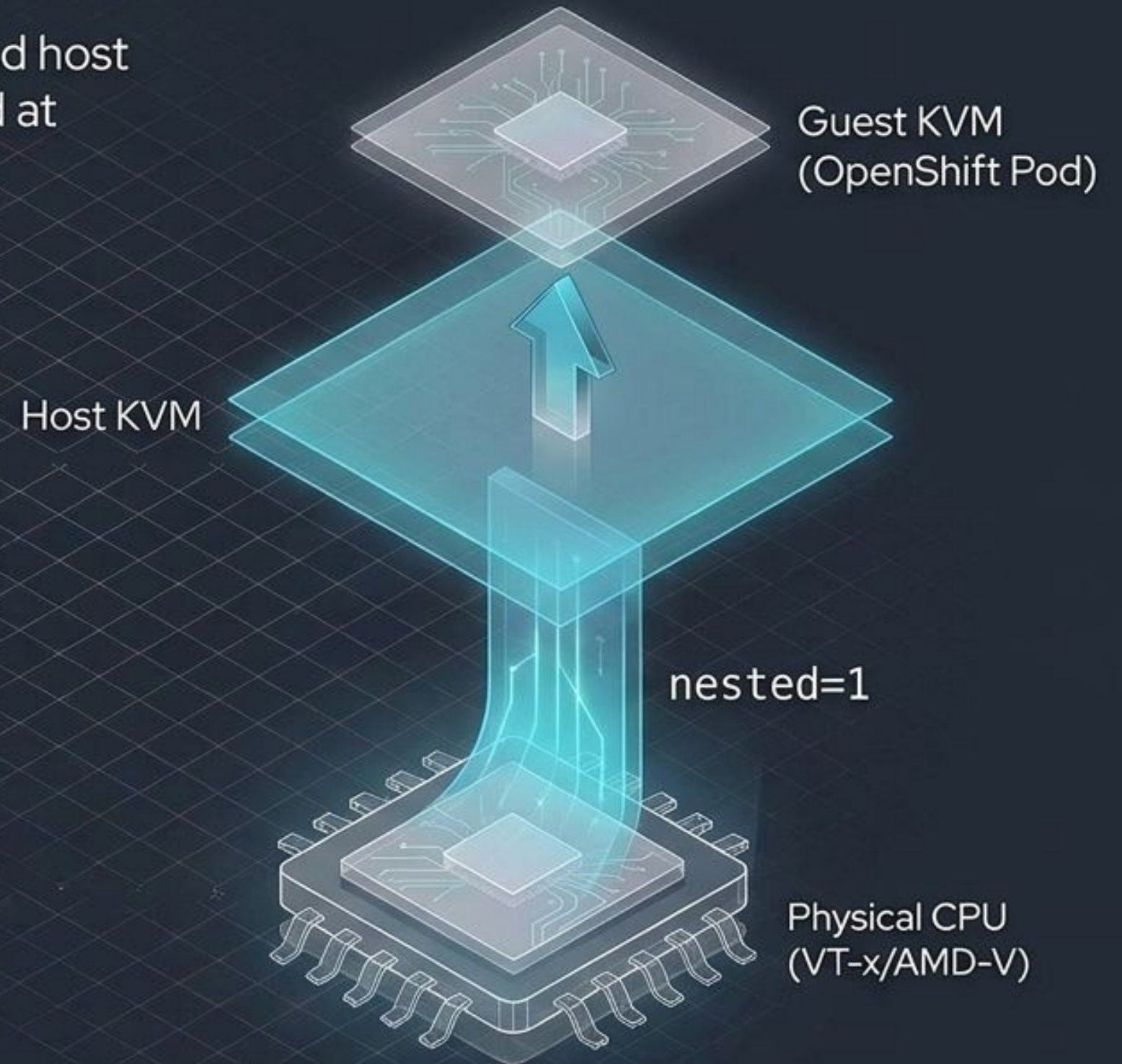
```
root@laptop:~# fdisk /dev/nvme0n1
```

```
Command (m for help): n
Partition type: primary
First sector: default
Last sector: +500G
```

```
root@laptop:~# dnf update -y
```


Bridging the Virtualization Extensions

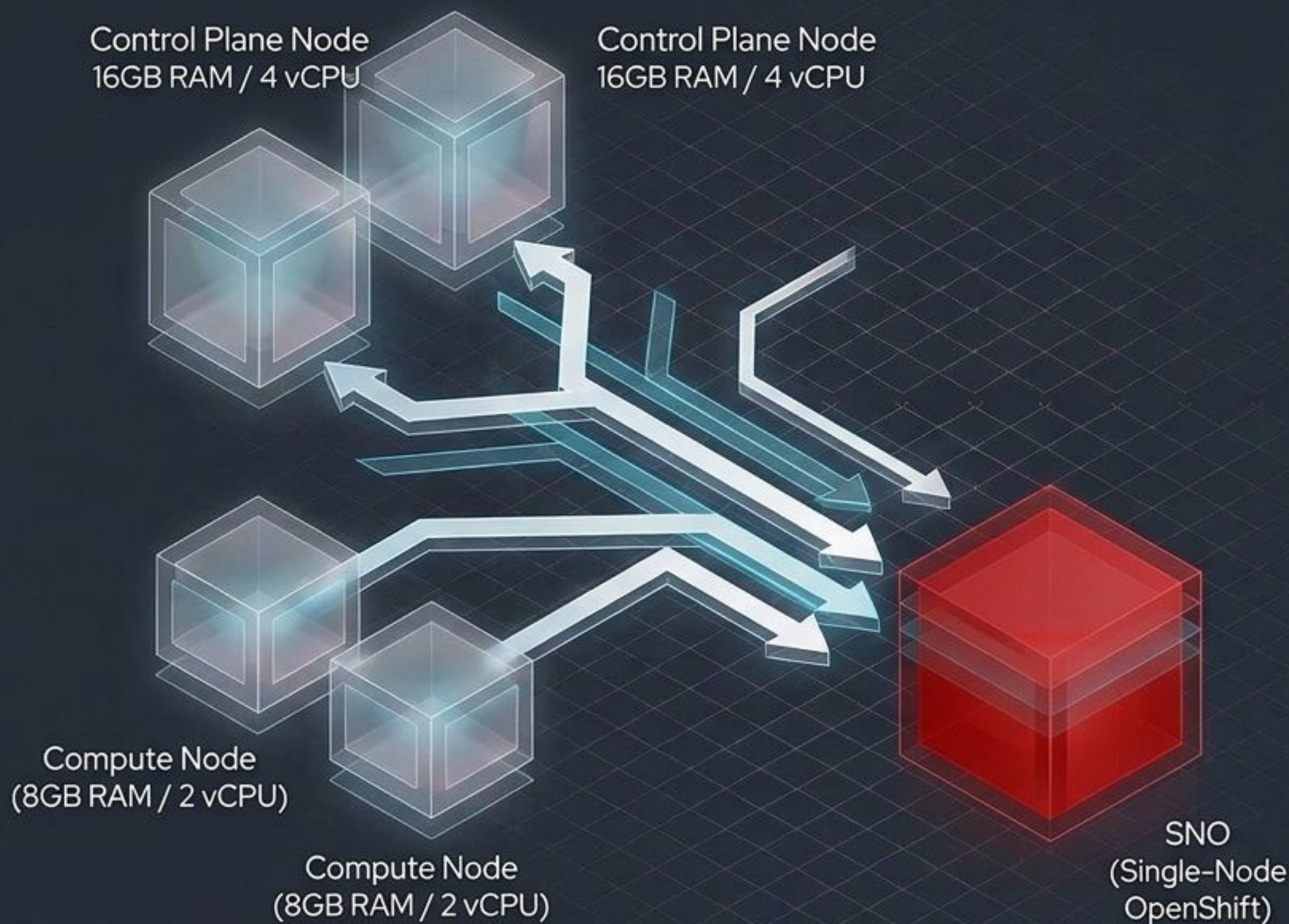
To allow a virtual machine to act as a hypervisor and host its own VMs, nested virtualization must be enabled at the host kernel level.



```
$ cat /sys/module/kvm_intel/parameters/nested  
1
```

**Note: For AMD systems, use `kvm_amd`.*

Layer 2: Single-Node OpenShift (SNO)



Explanatory Details:

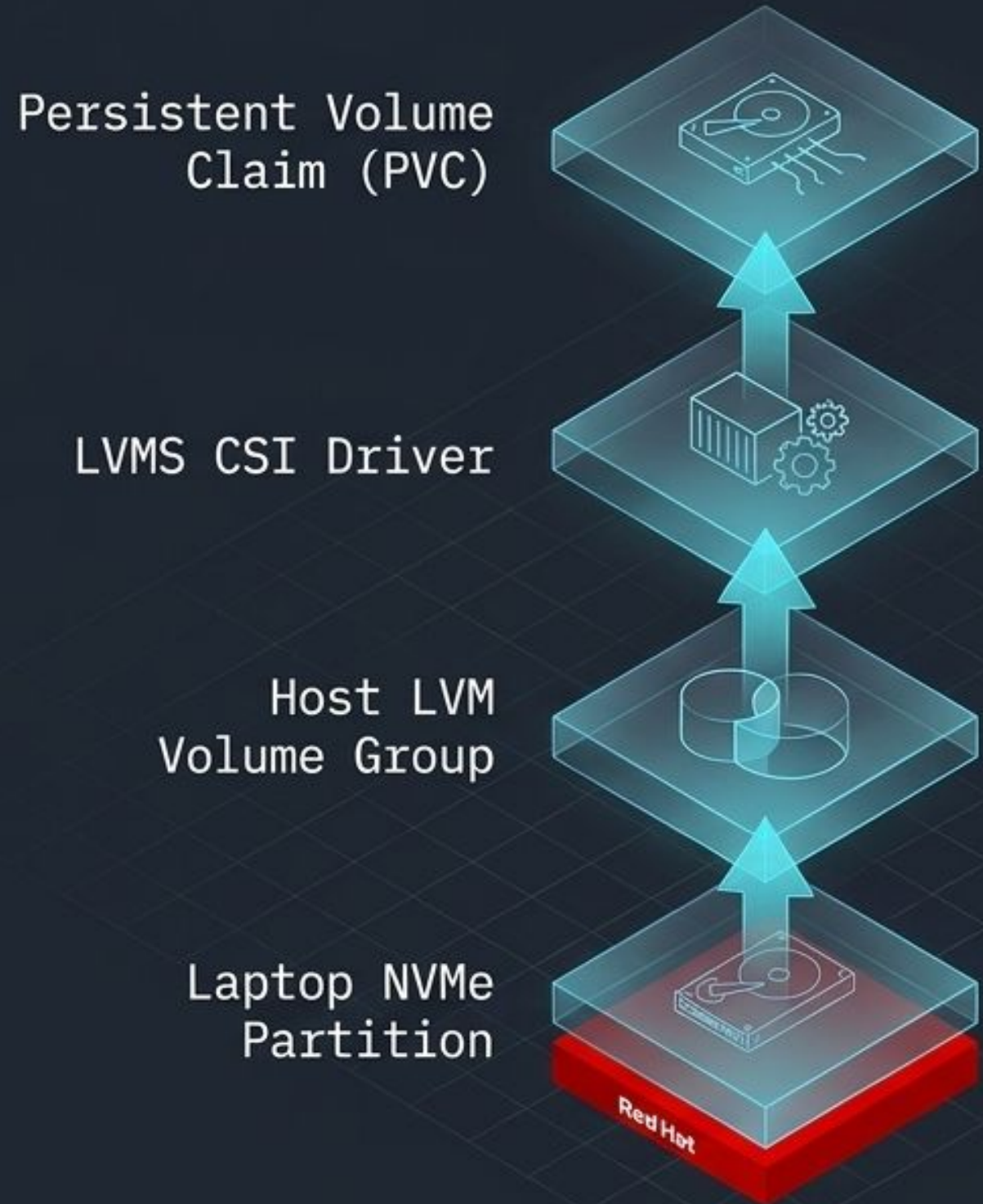
- **Base OS:** Red Hat Enterprise Linux CoreOS (RHCOS). Integrates CRI-O runtime and dedicated container tools.
- **Compression:** Combines Control Plane (normally 16GB RAM / 4 VCPU) and Compute (normally 8GB RAM / 2 vCPU) into a single, unified footprint.
- **Mechanism:** Uses rpm-ostree managed through the Machine Config Operator for declarative configuration.

Storage Engine Selection: LVMS vs. ODF

Feature/Constraint	OpenShift Data Foundation (ODF)	LVM Storage Operator (LVMS)
Architecture	Distributed Ceph cluster	Local volume manager on raw disk
Target Use Case	Highly Available multi-node clusters	Single Node OpenShift (SNO) and Edge deployments
Resource Overhead	High RAM/CPU cost for Ceph replication	Extremely lightweight, near-zero overhead
Access Modes	Native RWX (ReadWriteMany) support	RWO (ReadWriteOnce) focus

Conclusion: LVMS is the mandatory choice to preserve laptop RAM and CPU for the actual nested VM workloads.

Layer 3: Dynamic Local Storage Provisioning

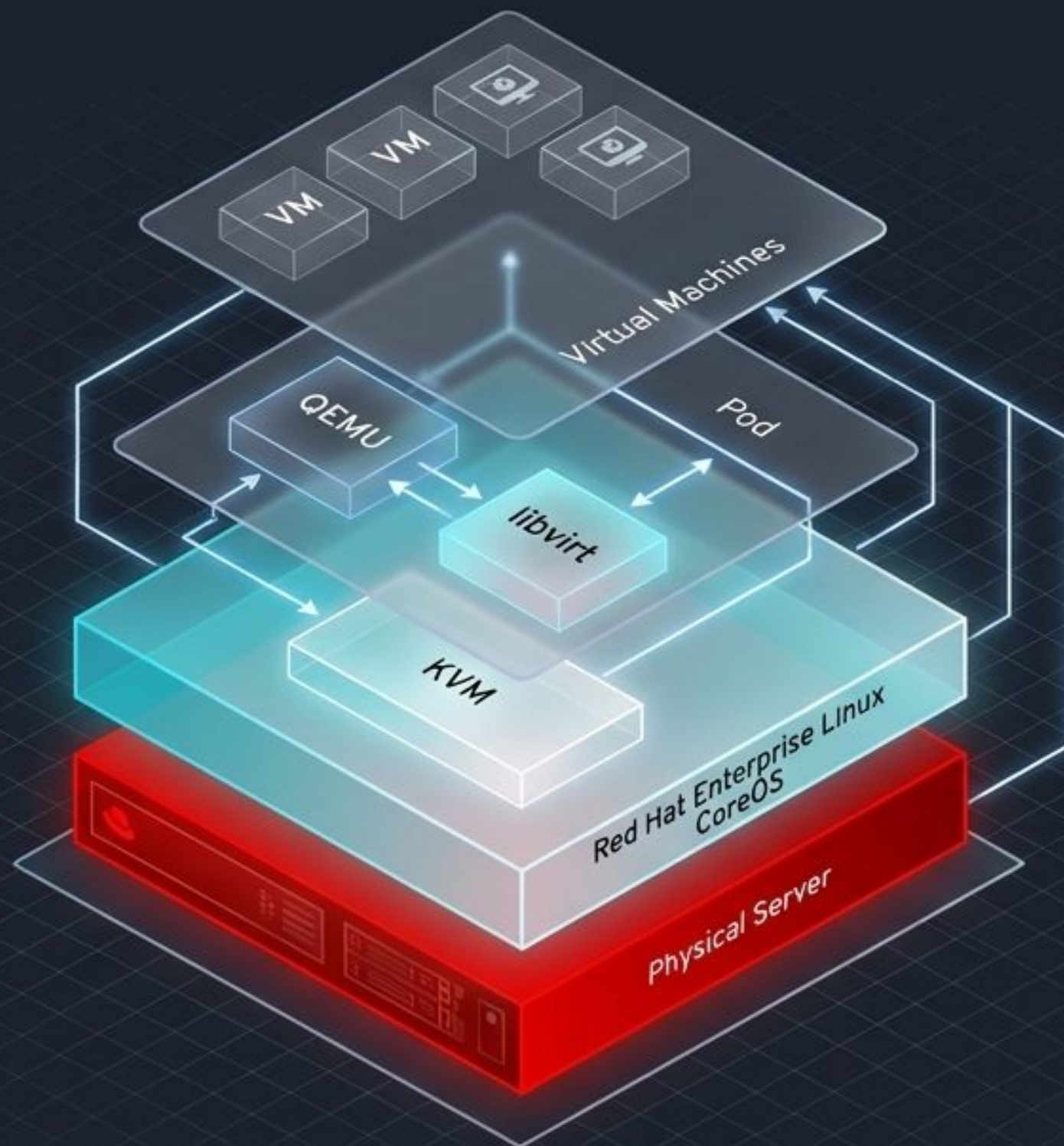


OpenShift Virtualization treats each virtual disk as an independent, managed resource via a Persistent Volume Claim (PVC).

The Container Storage Interface (CSI) coordinates the creation of the volume on the backing device and mounts it directly to the node hosting the VM.

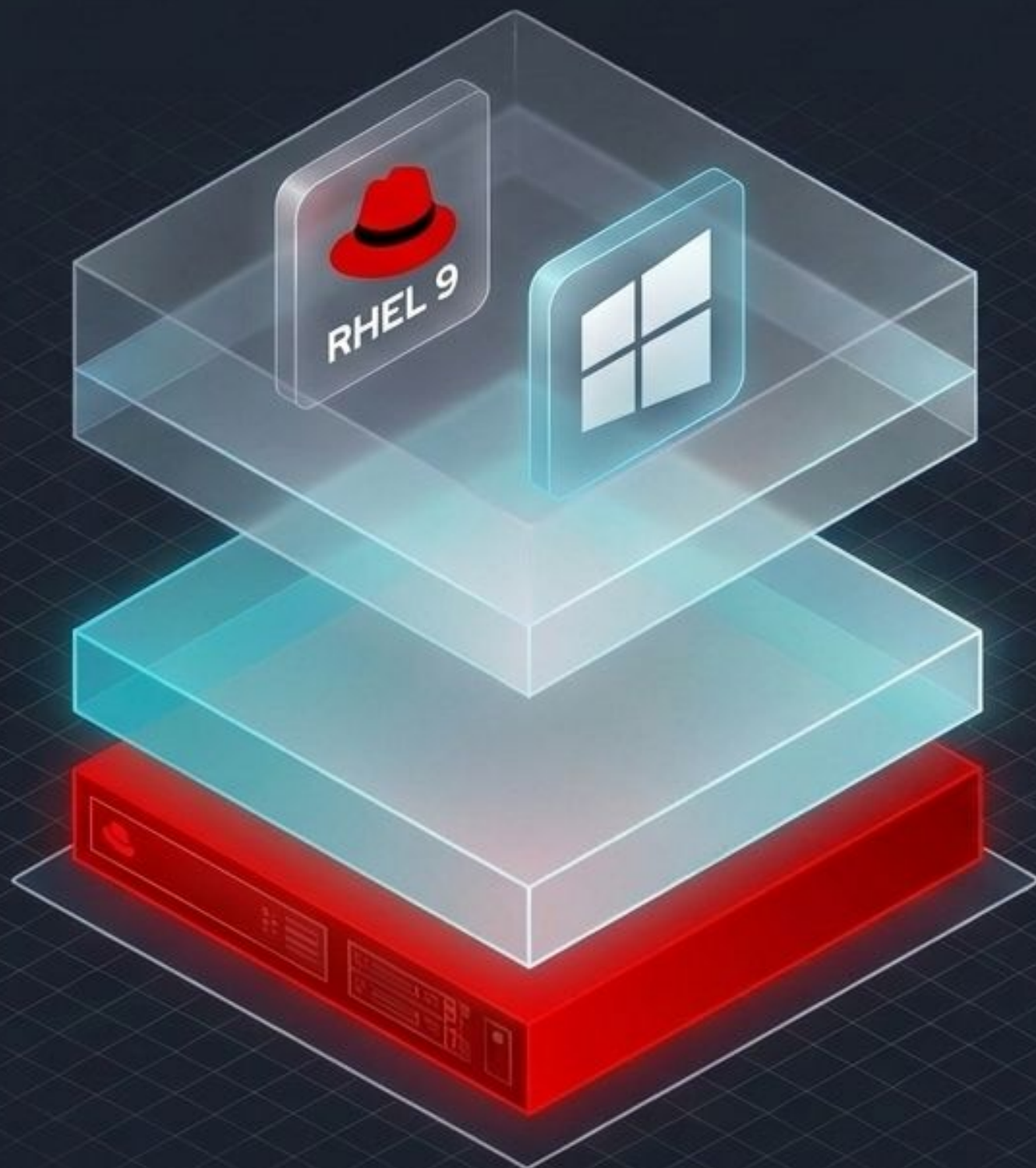
Because the VMs reside on a single node (the laptop), ReadWriteOnce (RWO) access mode is utilized instead of the RWX mode normally required for live migration.

Layer 4: OpenShift Virtualization (CNV)



- KVM is the same robust hypervisor used in Red Hat Virtualization and OpenStack, now containerized.
- Managed natively via Kubernetes Custom Resources.

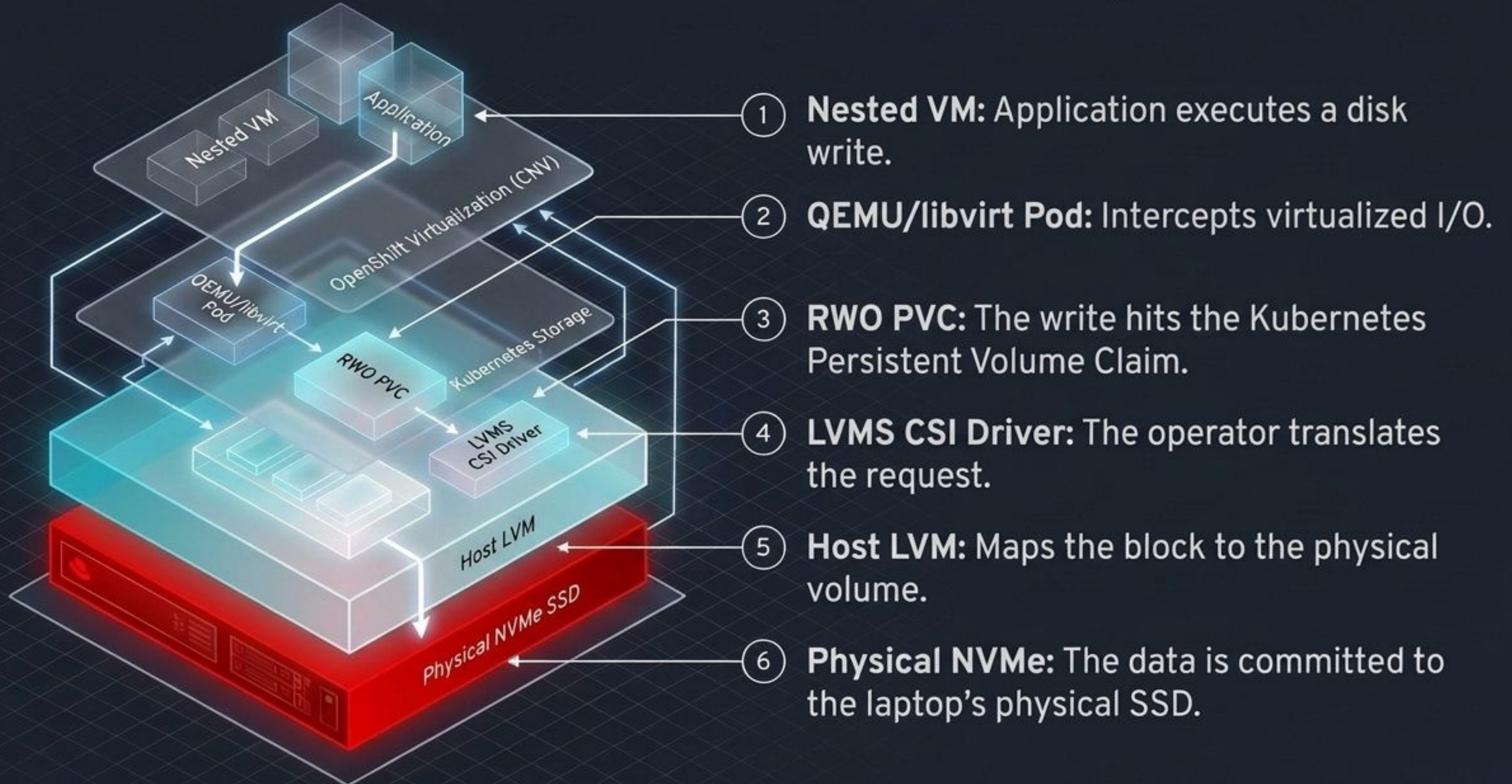
Layer 5: The Nested Virtual Machine Workload



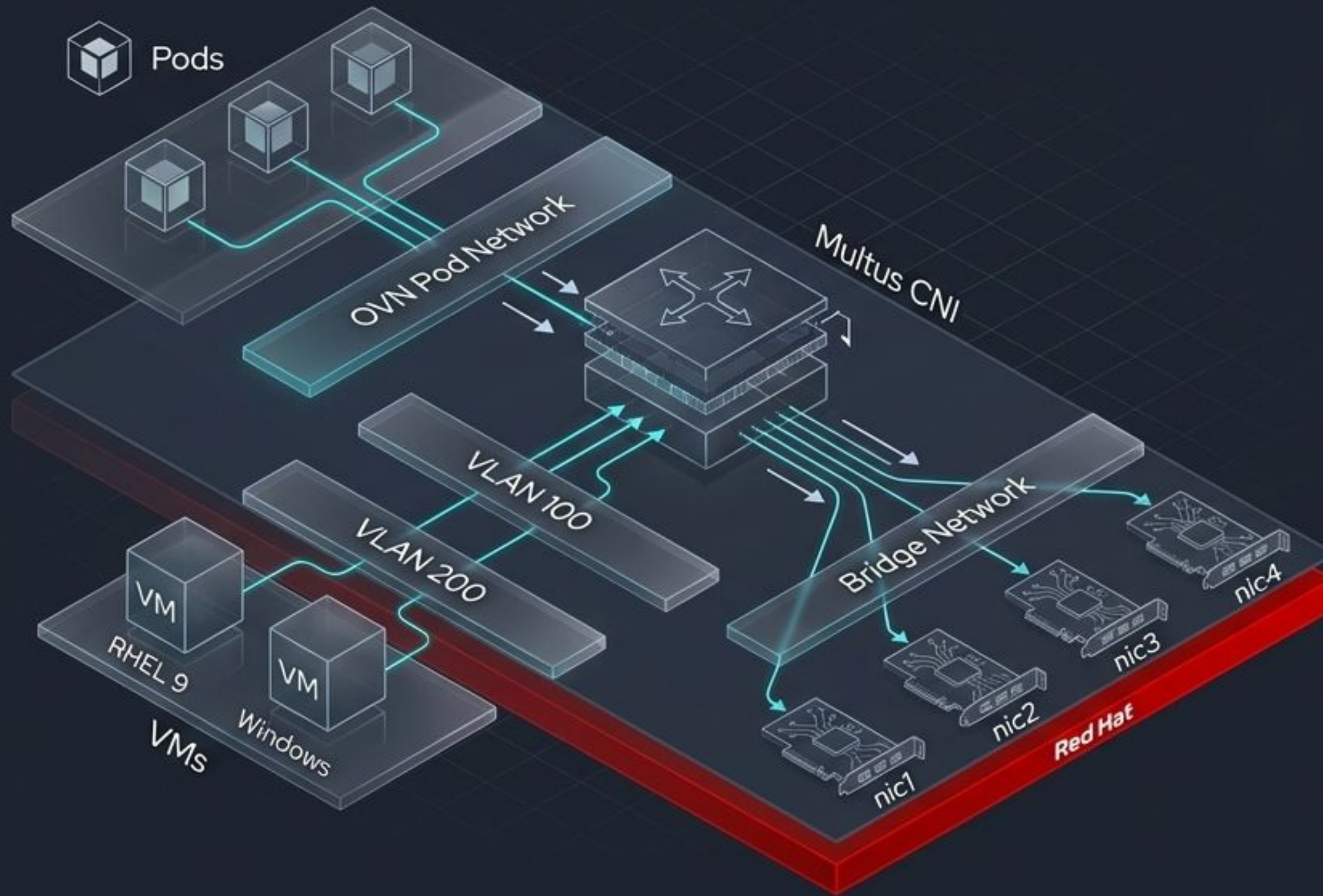
```
apiVersion: kubevirt.io/v1  
kind: VirtualMachine  
runStrategy: Always
```

- Virtual machines are native OpenShift objects.
- Can run alongside standard container pods within the same namespace.
- Supports full Linux distributions and Windows Subsystem for Linux (WSL) 2 (due to nested virt being enabled).

The Data Flow: Storage Routing

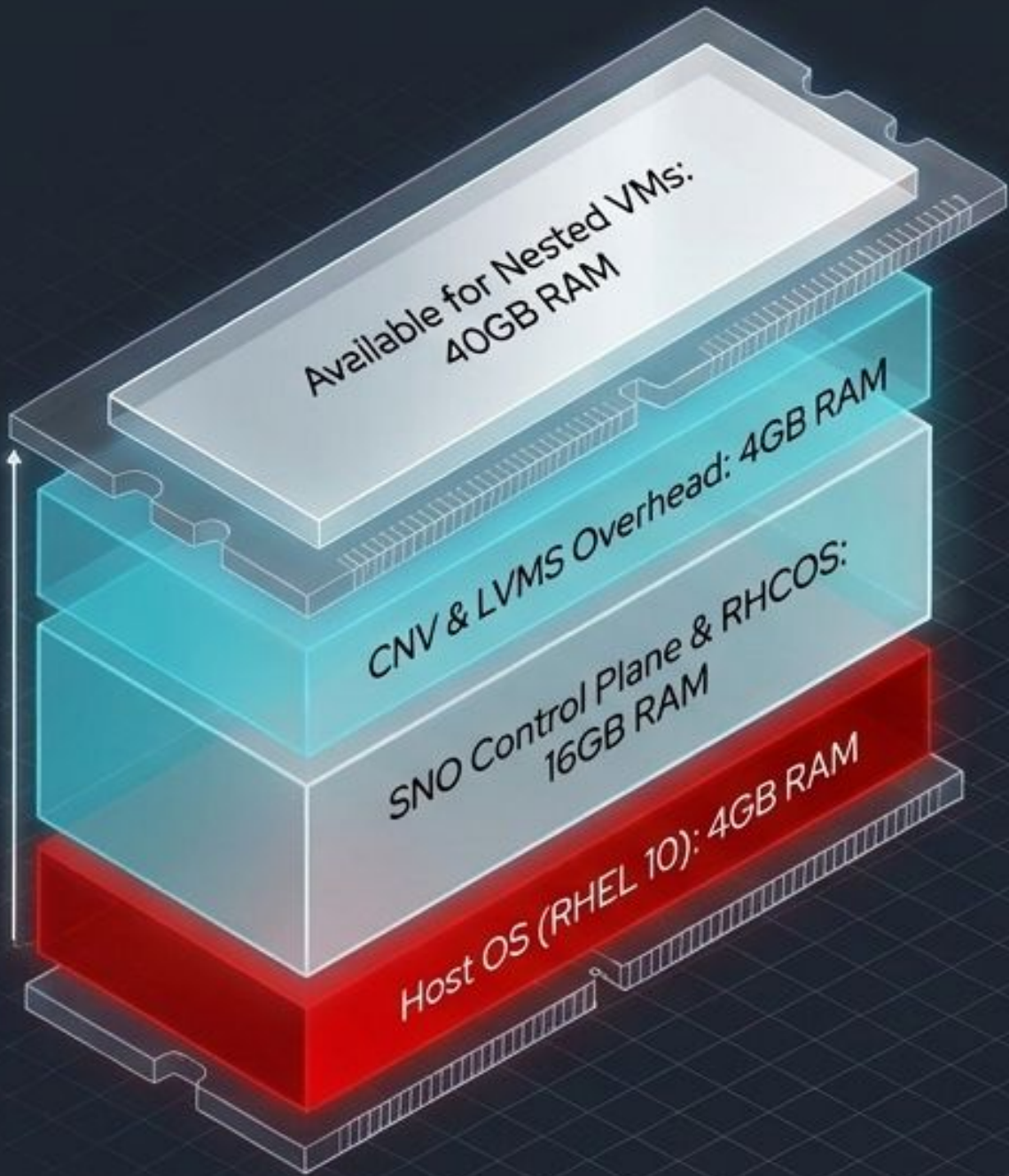


The Compute Flow: Network Routing



- **OVN-Kubernetes:** Provides the default pod networking overlay.
- **Multus CNI:** Allows the nested VM to attach to multiple network interfaces simultaneously.
- **Bridge Network:** Routes traffic from the VM's vNIC directly out through the laptop's physical Wi-Fi or Ethernet connection.

Resource Sizing for a 64GB Laptop Budget



Component	RAM Allocation	vCPU Allocation	Purpose
Host OS (RHEL 10)	4GB RAM	2 vCPUs	Base system overhead
SNO Control Plane & RHCOS	16GB RAM	4 vCPUs	Kubernetes API, ETCD, Operators
CNV & LVMS Overhead	4GB RAM	1 vCPU	Storage CSI, libvirt controllers
Available for Nested VMs	40GB RAM	Remaining vCPUs	Overcommit allowed

Takeaway: This precise tuning guarantees cluster stability while leaving over 60% of laptop memory available for the actual demo workloads.

Security & Hardening Trade-offs



Production Hardening - Level 1 Profile

- Nested virtualization is strictly disabled by default.
- Reduces attack surface to prevent malicious guests from accessing the hypervisor stack.



Demo Environment Reality

- Nested virtualization must be explicitly enabled.
- Required for VMs to utilize hardware virtualization to start their own VMs, or to run advanced guests like Windows with WSL2.
- This is an accepted, isolated risk for a closed-loop demo environment.

The Portable Datacenter Realized



- **No Compromises:** Run a full enterprise hybrid-cloud stack offline.
- **Architectural Purity:** Same APIs, same CSI drivers, same KVM technology as a 100-node bare-metal cluster.
- **Art of the Possible:** Prove the capabilities of RHEL 10 and OpenShift Virtualization anywhere, on a single machine.

Deploy the Blueprint.
Boot the Cluster. Run the Demo.